

MTH 204 : Lecture 10

Ex: $y'' + 4y = x \sin(2x) + 8$

(Note that here we need to use multiplication rule, modification rule and sum rule)

The characteristic Equation for the Homogeneous CDE

$$\lambda^2 + 4 = 0 \Rightarrow \lambda = \pm 2i$$

$$y_h(x) = c_1 \cos(2x) + c_2 \sin(2x)$$

$$y_p(x) = (A_1 x + B_1) (c_1 x \cos 2x + c_2 x \sin 2x) + E$$

$$\Rightarrow y_p(x) = A x^2 \cos 2x + B x^2 \sin 2x + C x \cos 2x + D x \sin 2x + E$$

Then $y_p'(x) = 2Ax \cos 2x - 2Ax^2 \sin 2x + 2Bx \sin 2x + 2Bx^2 \cos 2x$
 $+ C \cos 2x - 2Cx \sin 2x + D \sin 2x + 2Dx \cos 2x$

$$\Rightarrow y_p'(x) = 2Bx^2 \cos 2x - 2Ax^2 \sin 2x + (2A + 2D)x \cos 2x$$

$$+ (2B - 2C)x \sin 2x + C \cos 2x + D \sin 2x$$

$$\Rightarrow y_p''(x) = 4Bx \cos 2x - 4Bx^2 \sin 2x - 4Ax \sin 2x - 4Ax^2 \cos 2x$$

$$+ (2A + 2D) \cos 2x - (4A + 4D)x \sin 2x$$

$$+ (2B - 2C) \sin 2x + (4B - 4C)x \cos 2x$$

$$- 2C \sin 2x + 2D \cos 2x$$

Substituting and equating the coefficients :

$$x^2 \cos 2x \quad -4A + 4A = 0$$

$$x^2 \sin 2x \quad -4B + 4B = 0$$

$$x \cos 2x \quad 4B + 4B - 4C + 4C = 0 \Rightarrow 8B = 0 \Rightarrow \boxed{B = 0}$$

$$x \sin 2x \quad -4A - 4A - 4D + 4D = 1 \Rightarrow -8A = 1 \Rightarrow \boxed{A = -\frac{1}{8}}$$

$$\cos 2x : \quad 2A + 2D + 2D = 0 \Rightarrow 2A + 4D = 0 \Rightarrow D = -\frac{1}{2}A \\ \Rightarrow \boxed{D = \frac{1}{16}}$$

$$\sin 2x : \quad 2B - 2C - 2C = 0 \Rightarrow -4C = 0 \Rightarrow C = 0$$

$$\underline{\text{constant}} : 4E = 8 \Rightarrow \boxed{E = 2}$$

$$\text{So, } y_p(x) = -\frac{1}{8} x^2 \cos 2x + \frac{1}{16} x \sin 2x + 2$$

The general solution is

$$y(x) = C_1 \cos 2x + C_2 \sin 2x - \frac{1}{8} x^2 \cos 2x \\ + \frac{1}{16} x \sin 2x + 2$$

Existence and Uniqueness of Solutions:

Existence and Uniqueness Theorem for IVP:

If $p(x)$ and $q(x)$ are continuous functions on some open interval I and $x_0 \in I$, then the initial value problem

$$y'' + p(x)y' + q(x)y = 0$$

with two initial conditions $y(x_0) = K_0$, $y'(x_0) = K_1$
with given K_0 and K_1

has a unique solution $y(x)$ on the interval I

We will also give a theorem about existence of general solution.

Linear Independence of Solutions:

Consider the second order ODE: $y'' + p(x)y' + q(x)y = 0$
where $p(x)$ and $q(x)$ are continuous functions on an open interval I .

Two solutions y_1 and y_2 are called Linearly independent on I if the equation

$$C_1 y_1(x) + C_2 y_2(x) = 0 \text{ on } I \Rightarrow C_1 = C_2 = 0$$

We call y_1 and y_2 linearly dependent on I if this equation also holds for constants C_1 and C_2 not both zero.

y_1 and y_2 are linearly dependent iff y_1 and y_2 are proportional on I

(That is, $y_1 = k y_2$ or $y_2 = l y_1$ $\forall x \in I$ where k, l are constants.)

Theorem: Let us consider the ODE

$$y'' + p(x)y' + q(x)y = 0 \text{ where}$$

$p(x)$ and $q(x)$ are continuous functions on an open interval I .

Then two solutions y_1 and y_2 of the ODE on I are linearly dependent on I iff

$$\text{their Wronskian } W(y_1, y_2) = y_1 y_2' - y_1' y_2$$

is zero at some $x_0 \in I$

Furthermore if $W = 0$ at an $x = x_0 \in I$, then $W = 0$ on I ,

Hence, if there is an $x_1 \in I$ at which W is not 0, then y_1 and y_2 are linearly independent on I .

Proof:

$\Rightarrow$: Let y_1 and y_2 be linearly dependent on I

Then either $y_1 = ky_2$ or $y_2 = ly_1$ where k and l are constants.
Suppose $y_1 = ky_2$

$$W(y_1, y_2) = y_1 y_2' - y_1' y_2 = k y_2 y_2' - k y_2' y_2 = 0 \text{ on } I.$$

$$\Rightarrow W \equiv 0 \text{ on } I.$$

Similarly for $y_2 = ly_1$.

$\Leftarrow$: Conversely suppose $W(y_1, y_2) = 0$ for some $x_0 \in I$.

Let us consider the system of equation in unknown C_1 and C_2

$$\left. \begin{aligned} C_1 y_1(x_0) + C_2 y_2(x_0) &= 0 \\ C_1 y_1'(x_0) + C_2 y_2'(x_0) &= 0 \end{aligned} \right\}$$

$$\Rightarrow \begin{pmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Since $W(y_1(x_0), y_2(x_0)) = 0$, the above system has a non trivial solution $\{c_1, c_2\}$

$$\text{Let } Y(x) = \tau_1 y_1(x) + \tau_2 y_2(x)$$

By superposition principle $Y(x)$ is a solution of the ODE.

$$\text{Moreover } Y(x_0) = 0 \text{ and } Y'(x_0) = 0$$

$$\left. \begin{array}{l} \text{Thus } Y(x) \text{ is a solution of the IVP} \\ y'' + p(x)y' + q(x)y = 0 \\ y(x_0) = 0 \text{ and } y'(x_0) = 0 \end{array} \right\}$$

But $y^* \equiv 0$ is also a solution of the above IVP.

So, by the uniqueness theorem

$$Y \equiv y^* \equiv 0$$

$$\Rightarrow \tau_1 y_1(x) + \tau_2 y_2(x) = 0 \text{ on } I$$

where τ_1 and τ_2 are not both zeros.

Hence y_1 and y_2 are linearly dependent.

If $W(y_1(x), y_2(x)) = 0$ for some $x_0 \in I$
then y_1 and y_2 are linearly dependent.

Then by the first part, $W \equiv 0$ on I

- Thus if $W(y_1(x), y_2(x)) \neq 0$ at some point $x_1 \in I$ then y_1 & y_2 are linearly independent.

Note:

$$W(y_1, y_2) = \left(\frac{y_2}{y_1} \right)' y_1^2 \quad (\text{if } y_1 \neq 0)$$

$$\text{or, } W(y_1, y_2) = - \left(\frac{y_1}{y_2} \right)' y_2^2 \quad (\text{if } y_2 \neq 0)$$

$$\bullet W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \quad \left(\begin{array}{l} \text{Wronski determinant} \\ \text{or Wronskian} \end{array} \right)$$

Existence of a General Solution:

Theorem: If $p(x)$ and $q(x)$ are continuous functions on an open interval I , then the ODE $y'' + p(x)y' + q(x)y = 0$ ----- (1)

has a general solution on I .

Proof: Using the first theorem, the ODE (1) has a solution $y_1(x)$

Satisfying $y_1(x_0) = 1, y_1'(x_0) = 0$

Thus for $x_0 \in I$ we are considering two IVP:

① $y'' + p(x)y' + q(x)y = 0$
with initial conditions
 $y(x_0) = 1, y'(x_0) = 0$

and

② $y'' + p(x)y' + q(x)y = 0$
with initial conditions
 $y(x_0) = 0, y'(x_0) = 1$

for some $x_0 \in I$

and a solution $y_2(x)$ satisfying

$$y_2(x_0) = 0, y_2'(x_0) = 1$$

Now the Wronskian of these two solutions at $x = x_0$,

$$\begin{aligned} W(y_1(x_0), y_2(x_0)) &= y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0) \\ &= 1 \times 1 - 0 \times 0 = 1 \neq 0. \end{aligned}$$

So, $y_1(x)$ and $y_2(x)$ are linearly independent on I .

Thus $y(x) = c_1 y_1(x) + c_2 y_2(x)$ with arbitrary c_1 & c_2 is a general solution of (1).

Theorem (A general solution includes all solutions):

Consider the ODE : $y'' + p(x)y' + q(x)y = 0 \dots\dots (1)$

where $p(x)$ and $q(x)$ are continuous functions on some open interval I .

Then every solution $Y(x)$ of (1) on I is of the form $Y(x) = c_1 y_1(x) + c_2 y_2(x)$

where y_1 and y_2 are any two linearly independent solutions of (1) on I and c_1, c_2 are suitable constants.

Note: Hence (1) doesn't have singular solutions (ie. solutions not obtainable from a general solution)

Proof: Let $Y(x)$ be any solution of (1) on I .

By the previous theorem, the ODE (1) has a general solution $Y(x) = c_1 y_1(x) + c_2 y_2(x)$ on the interval I .

Choose $x_0 \in I$ and consider the system of equations:

$$c_1 y_1(x_0) + c_2 y_2(x_0) = Y(x_0)$$

$$c_1 y_1'(x_0) + c_2 y_2'(x_0) = Y'(x_0)$$

$$\Rightarrow \begin{pmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} Y(x_0) \\ Y'(x_0) \end{pmatrix}$$

Since y_1, y_2 are linearly independent, their Wronskian at x_0 is not zero.

Hence there is a unique solution $c_1 = \tau_1$

and $c_2 = \tau_2$

With τ_1 and τ_2 ,

Then $y^*(x) = \tau_1 y_1(x) + \tau_2 y_2(x)$ is a particular solution of (i)

with the condition,

$$y^*(x_0) = \tau_1 y_1(x_0) + \tau_2 y_2(x_0) = Y(x_0)$$

$$y^{*'}(x_0) = \tau_1 y_1'(x_0) + \tau_2 y_2'(x_0) = Y'(x_0)$$

By the uniqueness of solution of IVP,

$$y^* \equiv Y \text{ on } I$$

$$[y^*(x) = Y(x) \quad \forall x \in I]$$

$$\text{So, } Y(x) = \tau_1 y_1(x) + \tau_2 y_2(x) \quad (\text{QED})$$

Ex: Find Wronskian. Show linear independence by using quotient and confirm it by the second theorem.

Given $y_1 = e^{-x} \cos \omega x$, $y_2 = e^{-x} \sin \omega x$.

• $\frac{y_2}{y_1} = \tan \omega x$ (not a constant)
So, y_1 and y_2 are linearly independent

• Wronskian $W(y_1, y_2) = \begin{vmatrix} e^{-x} \cos \omega x & e^{-x} \sin \omega x \\ -e^{-x} \cos \omega x - \omega e^{-x} \sin \omega x & -e^{-x} \sin \omega x + \omega e^{-x} \cos \omega x \end{vmatrix}$

$$= -e^{-2x} \sin \omega x \cos \omega x + \omega e^{-2x} \cos^2 \omega x + e^{-2x} \sin \omega x \cos \omega x + \omega e^{-2x} \sin^2 \omega x$$

$$= \omega e^{-2x}$$

The solutions are linearly independent iff $\omega \neq 0$
($\frac{y_2}{y_1} = \tan \omega x = 0$ if $\omega = 0 \Rightarrow y_2 = 0$ and then y_1 and y_2 are linearly dependent)

Note: A second order linear homogeneous ODE for which $e^{-x} \cos \omega x$ and $e^{-x} \sin \omega x$ are solutions is $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + (1 + \omega^2) y = 0$

(Because from the two functions we can write the characteristic equation)

$$\text{as: } (\lambda + 1 - i\omega)(\lambda + 1 + i\omega) = 0 \Rightarrow \lambda^2 + 2\lambda + (1 + \omega^2) = 0$$

Ex: $y'' - 2y' + y = 0$

characteristic Equation $\lambda^2 - 2\lambda + 1 = 0$

$$\Rightarrow (\lambda - 1)^2 = 0 \Rightarrow \lambda = 1, 1$$

So, the general solution: $y(x) = c_1 e^x + c_2 x e^x$

Wronskian of e^x and $x e^x$:

$$W(e^x, x e^x) = \begin{vmatrix} e^x & x e^x \\ e^x & e^x + x e^x \end{vmatrix} = e^{2x} + \cancel{x e^{2x}} - \cancel{x e^{2x}} = e^{2x} \neq 0$$

So, the two solutions e^x and $x e^x$ are linearly independent.

Ex: (a) Find a second order homogeneous linear ODE for which the given functions are solutions.

(b) Show linear independence by the Wronskian.

(c) Solve the initial value problem :

$$e^{-kx} \cos \pi x, e^{-kx} \sin \pi x$$

$$y(0) = 1, y'(0) = -k - \pi$$

(a) The characteristic equation is:

$$[\lambda - (-k + i\pi)][\lambda - (-k - i\pi)] = 0$$

$$\Rightarrow [(\lambda + k) + i\pi][(\lambda + k) - i\pi] = 0$$

$$\Rightarrow (\lambda + k)^2 - i^2 \pi^2 = 0 \Rightarrow (\lambda + k)^2 + \pi^2 = 0$$

$$\Rightarrow \lambda^2 + 2k\lambda + (k^2 + \pi^2) = 0$$

Hence the ODE is:

$$\frac{d^2 y}{dx^2} + 2k \frac{dy}{dx} + (k^2 + \pi^2)y = 0$$

(b) The Wronskian of $e^{-kx} \cos \pi x$ and $e^{-kx} \sin \pi x$ is:

$$W(e^{-kx} \cos \pi x, e^{-kx} \sin \pi x) = \begin{vmatrix} e^{-kx} \cos \pi x & e^{-kx} \sin \pi x \\ -k e^{-kx} \cos \pi x & -k e^{-kx} \sin \pi x \\ -\pi e^{-kx} \sin \pi x & +\pi e^{-kx} \cos \pi x \end{vmatrix}$$

$$= \cancel{-k e^{-2kx} \cos \pi x \sin \pi x} + \pi e^{-2kx} \cos^2 \pi x + \cancel{k e^{-2kx} \sin \pi x \cos \pi x} + \pi e^{-2kx} \sin^2 \pi x$$

$$= \pi e^{-2kx} (\cos^2 \pi x + \sin^2 \pi x) = \pi e^{-2kx} \neq 0$$

Hence $e^{-kx} \cos \pi x$ and $e^{-kx} \sin \pi x$ are linearly independent.

(c) The general solution of the ODE will be

$$y(x) = c_1 e^{-kx} \cos \pi x + c_2 e^{-kx} \sin \pi x$$

$$y'(x) = c_1 (-k e^{-kx} \cos \pi x - \pi e^{-kx} \sin \pi x) \\ + c_2 (-k e^{-kx} \sin \pi x + \pi e^{-kx} \cos \pi x)$$

$$\text{Now } y(0) = 1 \Rightarrow c_1 + c_2 \times 0 = 1 \Rightarrow \boxed{c_1 = 1}$$

$$y'(0) = -k - \pi \Rightarrow c_1(-k) + c_2 \pi = -k - \pi$$

$$\Rightarrow -\cancel{k} + c_2 \pi = -\cancel{k} - \pi \quad (\text{Since } c_1 = 1)$$

$$\Rightarrow c_2 \pi = -\pi \Rightarrow \boxed{c_2 = -1}$$

So, the solution of the initial value problem is:

$$y = e^{-kx} \cos \pi x - e^{-kx} \sin \pi x$$

$$\Rightarrow \boxed{y = e^{-kx} (\cos \pi x - \sin \pi x)}$$

Ex: (a) Find a second order homogeneous ODE whose solutions are x^2 and $x^2 \ln x$

(b) Show linear independence by the Wronskian.

(c) Solve the initial value problem:

$$y(1) = 4, \quad y'(1) = 6$$

(a) $m=2$ is a real double root of the Auxiliary Equation (corresponding to Euler-Cauchy equation)

The auxiliary equation is $(m-2)^2 = 0$

$$\Rightarrow m^2 - 4m + 4 = 0$$

$$\Rightarrow m(m-1) - 3m + 4 = 0$$

Hence the ODE is $x^2 y'' - 3x y' + 4y = 0$

(b) Now Wronskian of x^2 and $x^2 \ln x$ is:

$$W(x^2, x^2 \ln x) = \begin{vmatrix} x^2 & x^2 \ln x \\ 2x & 2x \ln x + \frac{x^2}{x} \end{vmatrix}$$

$$= \cancel{2x^3 \ln x} + x^3 - \cancel{2x^3 \ln x} = x^3 \neq 0 \text{ for } x \neq 0$$

Hence the two solutions x^2 and $x^2 \ln x$ are linearly independent and the general solution of the ODE is:

$$y(x) = (c_1 + c_2 \ln x) x^2 = c_1 x^2 + c_2 x^2 \ln x$$

(c) Now $y(x) = c_1 x^2 + c_2 x^2 \ln x$

$$\Rightarrow y'(x) = 2c_1 x + 2c_2 x \ln x + c_2 x^2 \frac{1}{x}$$

$$\Rightarrow y'(x) = 2c_1 x + 2c_2 x \ln x + c_2 x$$

$$\text{Now } y(1)=4 \Rightarrow c_1 + 0 = 4 \Rightarrow c_1 = \boxed{4}$$

$$y'(1)=6 \Rightarrow 2c_1 + c_2 = 6$$

$$\Rightarrow 8 + c_2 = 6 \Rightarrow c_2 = \boxed{-2}$$

Therefore the solution to the initial value problem is :

$$\boxed{y(x) = (4 - 2\ln x)x^2}$$